

ICB-2152
Examination –Dec- 2023
B.Tech. V Sem : ICB
Software Engineering

Max. Marks : 70

Min. Marks : 22

Time : 3 Hrs

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a – Max 50 words,

Part b – Max 50 words,

Part c – Max 100 words and

Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
			02	I	1
Q.1	(a)	Write the difference between product, process, and project.	02	II	1
	(b)	Write different characteristics of software compared to hardware.	03	II	1
	(c)	Explain Software myths. Write different types of software myths.	07	V	1
	(d)	When we choose spiral model? Compare spiral model with RAD model.			
OR					
		Explain CMMI model for process improvement.	07	II	1
Q.2	(a)	Illustrate Requirement Engineering with a help of neat diagram.	02	II	2
	(b)	What is software requirement specification?	02	III	2
	(c)	Compare Functional and non-functional requirements.	03	III	2
	(d)	Design SRS for Hotel Management system.	07	V	2
OR					
		Why requirements elicitation is considered as a critical task in requirements engineering? Explain any two methods for requirements elicitation.	07	III	2
Q.3	(a)	What is Data Dictionary? Discuss its advantages.	02	II	3
	(b)	Define Object Oriented design? Discuss its characteristics.	02	II	3
	(c)	Draw a zero level DFD for ATM machine.	03	V	3

(d)	Considering "Hospital Information system" as an example explain the following UML diagrams in detail with the help of neat sketches. i. Use case diagram ii. sequence diagram iii. Class diagram	07	V	3
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OR

	What is Cohesion and Coupling? Write different types of cohesion with a suitable example.	07	V	3
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Q.4	(a)	Differentiate between validation testing and verification testing. ✓	02	II	4
	(b)	Write different levels of testing. ✓	02	I	4
	(c)	Is it possible to estimate software size before coding? Justify your answer with suitable example.	03	III	4
	(d)	Compare and contrast between Black box testing and White box testing. ✓	07	II	4

OR

	Describe in detail the complete software maintenance process.	07	II	4
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Q.5	(a)	Discuss about Formal technical reviews. ✓	02	II	5
	(b)	Write the ISO 9000 quality standards. ✓	02	II	5
	(c)	Write the difference between reactive and proactive risk.	03	III	5
	(d)	Explain RMMM Plan in detail. ✓	07	II	5

OR

	Explain all levels of COCOMO model .Assume that the size of an organic software product has been estimated to be 32,000 lines of code. Determine the effort required to develop the effort required to develop the software product and the nominal development time.	07	IV	5
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Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

CSE-1864(A)
Examination – May-June - 2022
B.Tech. VI Sem: Computer Science & Engineering
Software Engineering

Time : 3 Hrs

Max. Marks : 70

Min. Marks : 22

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

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Part a – Max 50 words, Part b – Max 50 words,

Part c – Max 100 words and Part d – Max 400 words.

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- Q.1 (a) Compare between software process and software product. 02
 (b) How SEI CMMI helps to improve software development process? 02
 (c) Which process model leads to software reuse? Why? 03
 (d) When we choose spiral model? Compare spiral model with RAD model. 07

OR

Explain different types of software myths in detail. 07

- Q.2 (a) Explain all the information in data dictionary? 02
 (b) Differentiate data flow diagram and state transition diagram. 02
 (c) Elaborate all requirement analysis principles. 03
 (d) Illustrate Requirement Engineering with a help of neat diagram. 07

OR

Design 0 level and 1 level DFD for ATM machine. 07

- Q.3 (a) Explain the golden rules for interface design. 02
 (b) Justify "Design is not coding and coding is not design". 02
 (c) With the help of your own example compare Event flow diagram and Data flow diagram. 03
 (d) Considering "Hospital Information system" as an example explain the following UML diagrams in detail with the help of neat sketches. 07
 i) Use case diagram
 ii) Sequence diagram
 iii) Class diagram

OR

Explain data architectural and procedural design for software. 07

- Q.4 (a) What is the need for cyclomatic complexity?
(b) Explain unit testing.
(c) What is the importance of debugging and reliability analysis?
(d) Compare and contrast between Black box testing and White box testing.

OR

What are different types of software maintenance? Compare each of them.

- Q.5 (a) Explain the different categories of risks.
(b) What is meant by software review?
(c) "A good program has certain measurable attributes". Justify the statement.
(d) What do you mean by software quality assurance? How to system testing plays the role in software quality assurance. Justify your answer.

Explain RMMM plan with the help of suitable example.

MCA-202
Examination – June - 2022
MCA II Sem
Software Engineering

Time : 3 Hrs

Max. Marks : 70

Min. Marks : 28

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Word limit be observed as follows:

Part a – Max 50 words, Part b – Max 50 words,

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Word limit NOT to be followed for diagram, numerical, derivation.

- Q.1 (a) What do you understand by term 'system'? Explain. 02
- (b) Is Software engineering is necessary in software development process? Justify your answer. 02
- (c) Discuss the Concepts of MIS and DSS. How are they related? How do they differ? 03
- (d) What is prototype? Under what circumstances is it beneficial to construct prototype? Does construction of the prototype always increase the overall cost of the Software development? 07

OR

Make a comparative study of different life cycle process models? State the advantages and disadvantages of different process models? 07

- Q.2 (a) What is feasibility study? Explain its types and purpose. 02
- (b) "A Data dictionary is a structured repository of data about data". Discuss? 02
- (c) What is SRS? Discuss characteristics and components of the SRS. 03
- (d) Explain the statistical software quality assurance and also discuss software quality assurance activities. 07

OR

Describe the concept and procedure used in constructing DFDs. Draw the DFD for online shopping system. Make necessary assumptions for the same. 07

- Q.3 (a) What do you mean by term 'cohesion and coupling'? Discuss. 02
- (b) Is it possible to estimate software size before coding? Justify your answer with examples. 02
- (c) What effect is the project monitoring activity likely to have on the development process? 03
- (d) Discuss the role of cost estimation in a software development project. Briefly explain COCOMO model for cost estimation for all categories of projects? 07

OR

A project size of 200 KLOC is to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the effort, development time, average staff size and the productivity of the project. Considering $(a, b) = (3.0, 1.12)$ as multiplicative and exponential factor for the basic COCOMO effort estimation equation and $(c, d) = (2.5, 0.35)$ as multiplicative and exponential factor for the basic COCOMO development time estimation equation. 07

- Q.4 (a) Can software be correct and still not reliable? Justify your answer. 02
- (b) What do you understand by verification and validation? Differentiate with each other. 02
- (c) What do you understand by white box testing technique? Compare with black box testing technique. 03
- (d) Write a short notes on the following: 07
- (i) Regression Testing
 - (ii) Alpha Testing
 - (iii) Performance Testing

OR

Consider a simple program to classify a triangle. Its input is a triple of positive integers (say x, y, z) and the data type for input parameters ensures that these will be integers greater than 0 and less than or equal to 100. The program output may be one of the following words: Scalene, Isosceles, Equilateral, Not a triangle. Determine the equivalence class test suite for same. 07

- Q.5 (a) What are the main advantages of using CASE tools? 02
- (b) What is the difference between re-engineering and reverse engineering? 02
- (c) Why software maintenance is necessary? Give the importance of it. 03
- (d) Why it is important for an organization to undertake an effective reuse program? What are the important reuse artifacts that can be reused? Why is reuse of software components much more difficult than hardware components? 07

OR

What is the Software Configuration Management (SCM) process? Why it is needed in addition to the development process? 07

CS-1864

Examination –April/May- 2023

**B.Tech. VI Sem : Computer Science & Engineering
Software Engineering**

Time : 3 Hrs

Max. Marks : 70

Min. Marks : 22

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a – Max 50 words, Part b – Max 50 words,

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

- Q.1 (a) Write disadvantages of Waterfall model for software development. 02
 (b) What is a software process? Why is it important? 02
 (c) Briefly explain the role of management in software development. 03
 (d) Explain CMMI model for process improvement. 07

OR

Explain Spiral model in detail and write their advantages and disadvantages. 07

- Q.2 (a) What is Software requirement specification? 02
 (b) Explain functional and nonfunctional requirements. 02
 (c) Write the elements of the requirements engineering process. 03
 (d) Why requirements elicitation is considered as a critical task in requirements engineering? 07
 Explain any two methods for requirements elicitation.

OR

Write different characteristics of SRS. 07

- Q.3 (a) Differentiate between stamp coupling and content coupling. 02
 (b) What is the significance of adopting programming practices and coding standards? 02
 (c) What is 0 level DFD? Explain with suitable example. 03
 (d) What is the need of a modular system? Describe the effects of cohesion and coupling in modular design. 07

OR

What is Cohesion? Write different types of cohesion. 07

- Q.4 (a) What do you understand by the term system testing? 02
 (b) Explain two types of Black box testing strategies. 02
 (c) Write the difference between testing and debugging. 03

(d) What do you mean by maintenance? Write different types of maintenance.

07

OR

What do you mean by White box testing? Differentiate white box testing and black box testing.

07

Q.5 (a) Write difference between proactive and reactive risk.

02

(b) How does software prototyping help to increase the overall quality of the software?

02

(c) How ISO 9000 helps in software process improvement?

03

(d) What do you mean by software quality assurance? Explain it.

07

OR

What are risk management activities? Is it possible to prioritize risk

07

(d)	Explain CORBA and its Architecture.	OR	07
	Write a note on (Any Two)		
	(a) COM+		
	(b) DCOM		
	(c) EJB		
			

Total Printed Page : 02

Roll No. :

CS-1163
Examination- Nov/Dec - 2016
B.E.- VI- Sem: Computer Science and Engineering
Software Engineering

Time: 3 Hrs

Max. Marks: 70

Min. Marks: 22

Note: Total number of questions are 10. Attempt one question from each unit (including all parts). Assume missing data, if any, suitably.

UNIT-I

- Q1.(a) Write the difference between software product and software process. 07
 (b) Is CMM a software process model? Justify your answer by giving suitable example 07

'OR'

- Q2.(a) List the advantages and disadvantages of incremental model. 07
 (b) Compare and contrast spiral model with prototype model. 07

UNIT-II

- Q3.(a) Draw a Use-case Diagram for an Online Shopping System. 07
 (b) Write different characteristics of software requirement specification. 07

'OR'

- Q4.(a) Draw a level -1 DFD for ATM machine. 07
 (b) Write different requirement elicitation techniques. 07

UNIT-III

- Q5.(a) Describe the metrics for the design model of a product. What are the attributes of effective software metrics? 07
 (b) List the guidelines for good design. 07

'OR'

- Q6.(a) Describe the important principles and steps of user interface analysis and design. 07
 (b) Briefly explain what is meant by the terms modularity, cohesion and coupling within the context of software design and programming and then discuss their relevance. 07

UNIT-IV

- Q7.(a) Suppose a program contains 4 decision points, each of which has two branches. How many test cases are needed to perform path testing on such a program? Show clearly how you arrived at the answer. 07
 (b) What are the various types of system testing? 07

'OR'

- Q8.(a) Distinguish between black box and white box testing. Is both testing complementary to each other? 07

- (b) Briefly explain what is meant by the terms modularity, cohesion and coupling within the context of software design and programming and then discuss their relevance.

14 / 100



UNIT-IV

- Q7 (a) Suppose a program contains 4 decision points, each of which has two branches. How many test cases are needed to perform path testing on such a program? Show clearly how you arrived at the answer.
- (b) What are the various types of system testing?

'OR'

- Q8 (a) Distinguish between black box and white box testing. Is both testing complementary to each other?
- (b) What are the testing principles the software engineer must apply while performing the software testing?

UNIT-V

- Q9 (a) Draw a block diagram to illustrate the reengineering process.
- (b) Explain the scheme to handle changes in the software.

'OR'

- Q10 (a) The value of size in KLOC and different cost drivers for a project are given below.

Size = 200 KLOC

Cost driver

Software reliability = 1.15

Product complexity = 0.85

Execution time constraint = 1.00

Calculate the effort for three types of project i.e. Organic, Semidetached, Embedded, using COCOMO model.

- (b) Explain risk management. What types of Risks are involved in IT Projects?



Total Printed Pages: 03

Roll No.

CS/IT-1751

Examination – Nov.-Dec.- 2018

B.E. V Sem : CSE/IT

Software Engineering

Time : 3 Hrs

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Min. Marks : 22

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- Q.1 (a) Define Software Engineering 02
- (b) What is the role of risk management activity in spiral model. 02
- (c) Write short note on Project Scheduling 03
- (d) The value of size of program in KLOC and different cost drives are given below: 07
- Size-300KLOC, Complexity-0.95, Analyst Capability-1.05, Application of Software Engineering Method-0.8, performance Requirement-0.75.
- Calculate the effort for three types of projects i.e Organic, semidetached, embedded using COCOMO Model.

Explain RAD Model to achieve rapid development. How this model is different from linear sequential model. 07

- Q.2 (a) What is a requirement? Explain briefly. 02
 (b) What do you mean by Data dictionary 02
 (c) Write different requirement elicitation techniques. 03
 (d) What do you mean by feasibility study? Explain different types of feasibilities. 07

OR

Explain software requirement specification. What are the characteristics of good SRS? 07

- Q.3 (a) Write the guidelines for good design. 02
 (b) What is state transition diagram? 02
 (c) What is the use of drawing context diagram? 03
 (d) What is Modularity? List important properties of modular design. What problems are likely to arise if two modules have.
 (i) High coupling (ii) Low coupling 07

OR

What is user interface design? What do you know about GUI? 07

- Q.4 (a) What do you mean by coding? State its purpose. 02
 (b) Explain debugging. Write debugging process. 02

(2)

(c) What do you understand by testability? How can it be measured? 07

- (d) Differentiate between black box and white box testing. 07

OR

Explain cyclomatic complexity. How can it be calculated. 07

- Q.5 (a) Define the relationship between people and effort. 02
 (b) Explain the term scheduling and tracking. 02
 (c) What are measures, metrics and indicators? 03
 (d) Explain decomposition techniques. Differentiate between Loc based and FP based estimation. 07

OR

What is RMMM Plan? Explain in detail. 07

(3)

MCA-202
Examination – June - 2022
MCA II Sem
Software Engineering

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OR

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- (b) Is it possible to estimate software size before coding? Justify your answer with examples. 02
- (c) What effect is the project monitoring activity likely to have on the development process? 03
- (d) Discuss the role of cost estimation in a software development project. Briefly explain COCOMO model for cost estimation for all categories of projects? 07